

Reinforcement Learning in The Binding of Isaac

Purpose

The goal of our project is to create an autonomous agent in the game, The Binding of Isaac, that can navigate through different rooms and kill enemies using reinforcement learning. Our agent learns by interacting with the environment and getting rewarded for positive actions, such as avoiding damage and clearing rooms. The decision-making policy is improved by favoring actions that maximize rewards.

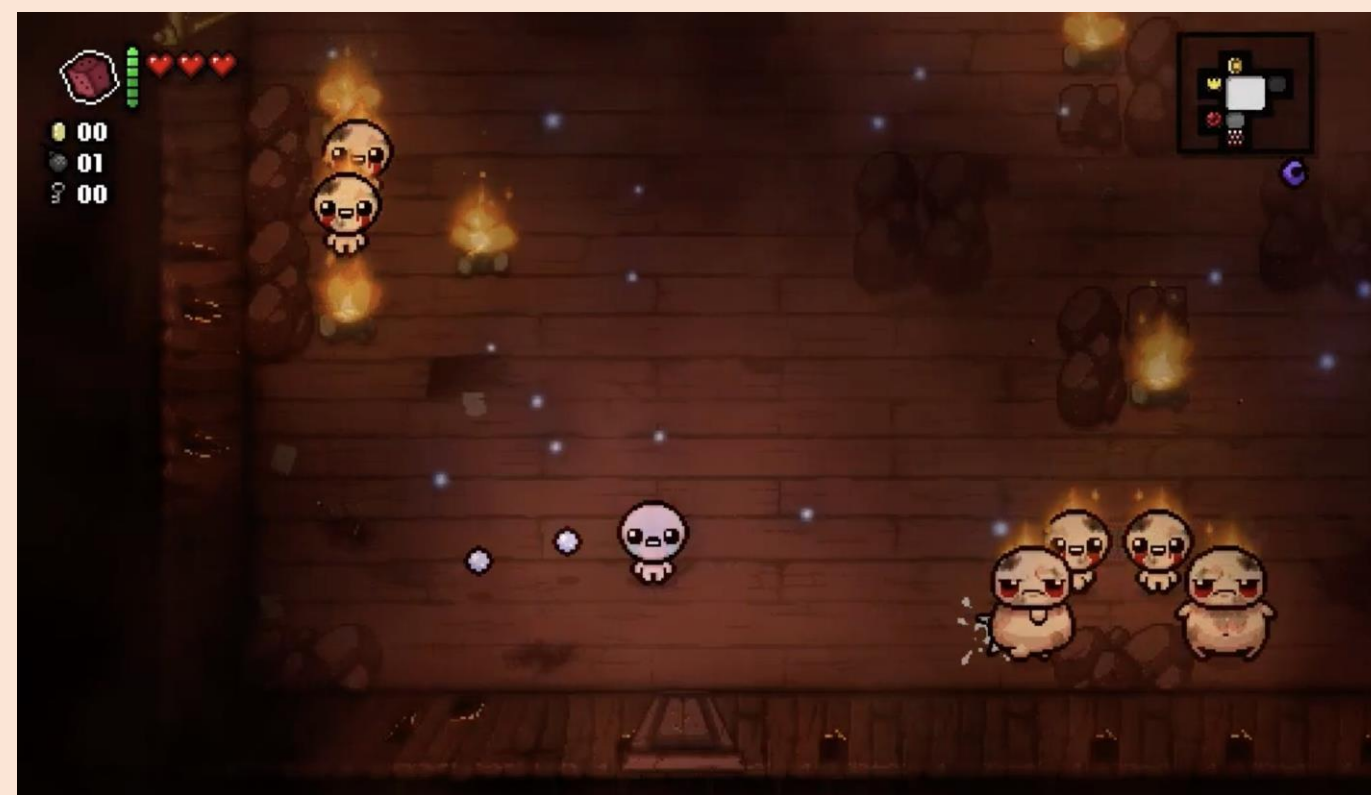


Figure 1. Example Game State Observed by the Agent. The agent receives a compact feature vector describing the tactical situation, such as enemy location, distances, health bar, etc.

Methods

We used a Deep Q Network (DQN) (see Figure 2) implemented in PyTorch, which estimates the long-term value of each action from the current game state. The agent interacted with the game environment through a custom Lua/REPENTOGON interface that handled the communication between the game and PyTorch. During training, each experience was stored in a replay buffer, and the network was updated by sampling past transitions. Over time, the DQN learned how to choose actions that improved the agent's survival and performance.

Figure 2. Deep Q-Network (DQN) architecture (source: GeekforGeeks, 2025). The network takes the current game state as input and outputs Q-values for each possible action, allowing the agent to estimate the long-term value of different behaviors during training.

After 500+ training episodes, the agent's performance greatly improved, as shown in Figure 4. The average reward increased from -846 (episodes 1-10) to -249 (episodes 528-537), with reduced variability, indicating more stable behavior. These results show that the agent learned to survive longer and take fewer penalties over time.

From observing the gameplay after training the model, it is able to actively dodge incoming projectiles and avoid touching enemies to avoid damage. However, shooting direction was still inaccurate.

Overall, this project shows that a deep reinforcement learning agent can successfully learn spatial navigation and survival tactics within a dynamic environment. The difference between the agent's proficiency in dodging versus its accuracy in shooting highlights the complications of training a joint action space. Since survival rewards were more immediate and easier for the model to earn rewards, the agent developed a "safety-first" strategy, prioritizing survival over aggressive combat.

Results and Conclusion

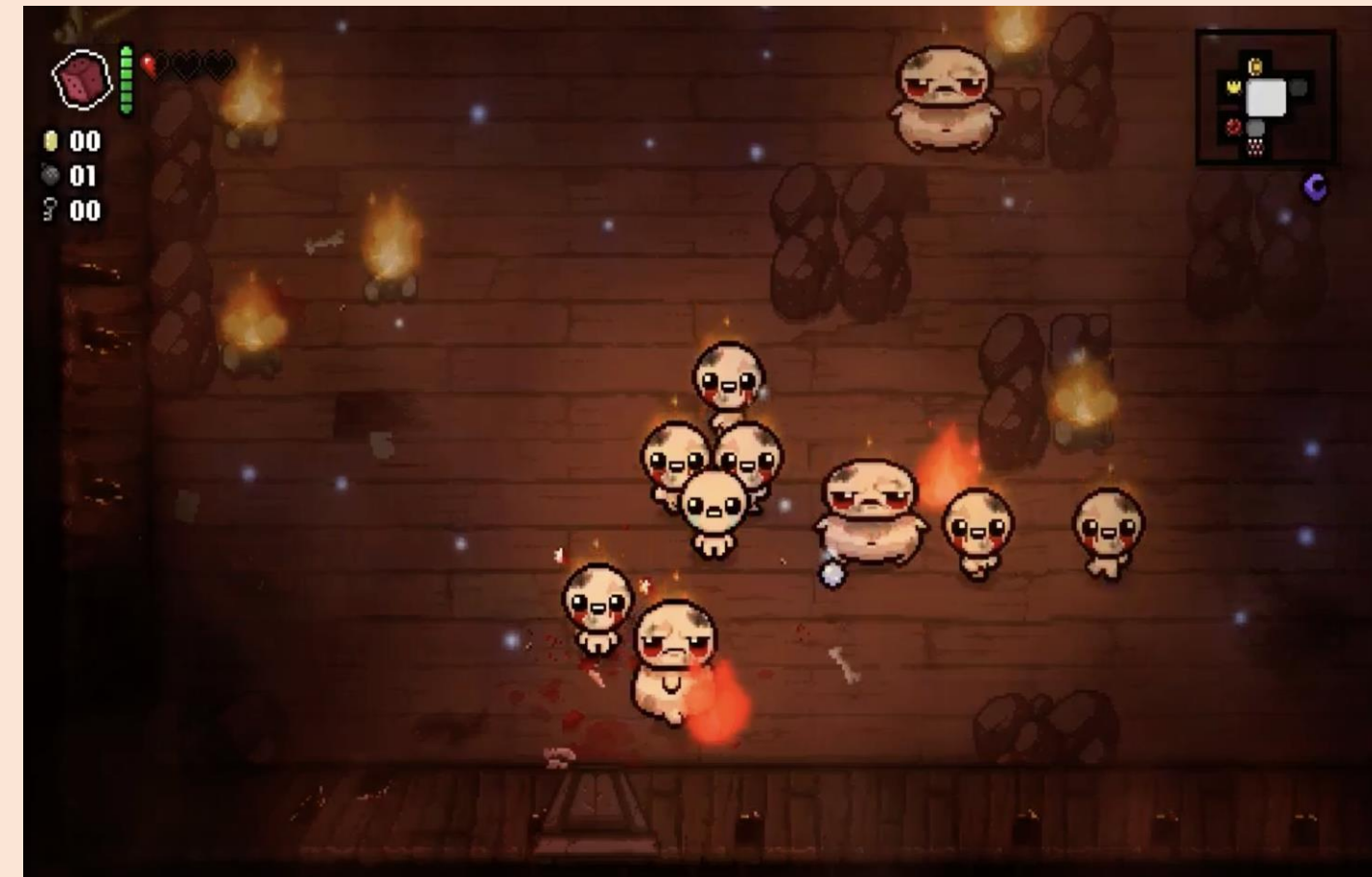


Figure 3. In-Game Screenshot Early Training. Without much training data, Isaac (positioned in the middle of the screenshot) tends to body-check into enemies, causing him to die sooner.

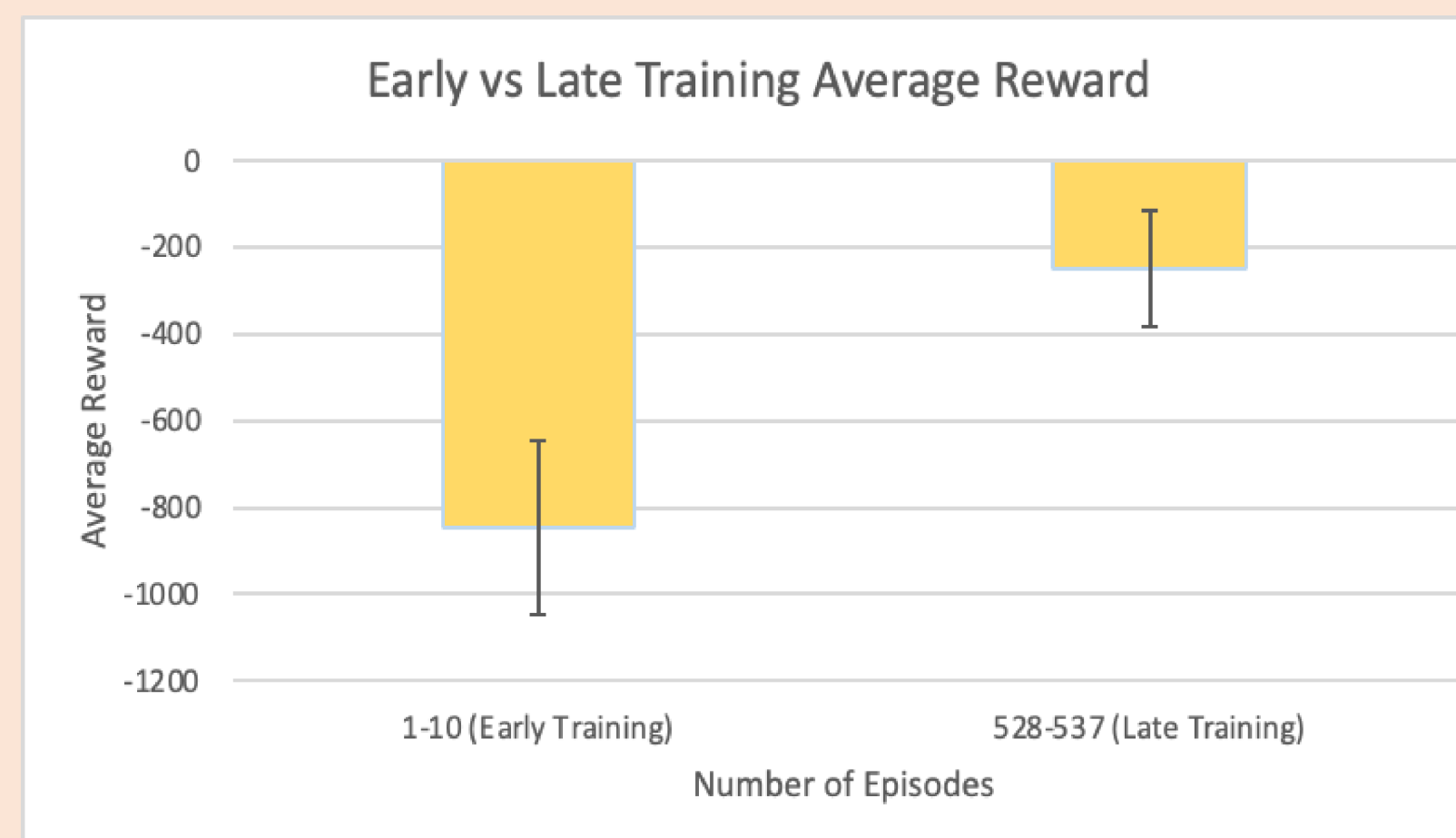


Figure 4. Early vs. Late Training Performance. Comparison of the agent's average reward (10-episode moving average) and variability between early episodes (1-10) and late episodes (528-537).

Limitations

Difficulty learning movement and shooting simultaneously

- Movement and hazard avoidance improved faster than aiming

Limited training signal for shooting accuracy

- Enemy HP drops observed, but unclear which shot caused damage

- Resulted in delayed and noisy reward signals

Not fully end-to-end learning

- Key decisions (navigation, safety) handled by rule-based modules
- Limits the agent's ability to learn all behaviors from experience

Future Work

Train model to navigate and explore independent of combat

- We successfully trained the model to handle combat in rooms, but navigation between rooms was still rule based (go through least used door). Potential future work would include training this independently, so that the model will be able to play purely using trained policies.

Model state space more optimally

- The current model relies on a "point-based" state space, tracking the two closest enemies and projectiles. While this is computationally efficient and effective for isolated encounters, it fails when there are multiple threats. Future iterations should explore more robust environmental modeling, to allow the model to perceive elements as a unified spatial context rather than a list of individual coordinates.

Decoupling Action Spaces for Specialized Learning

- To resolve the credit assignment problem, future work could decouple movement and shooting into two different specialized networks with independent reward signals. This approach would prevent survival-based rewards from overshadowing offensive training, allowing the agent to learn and improve on both aiming and defensive positioning.

References

Bol: Lua API Docs. (2024, December 9). <https://moddingofisaac.com/docs/rep/>

catinsured. (2025, November 29). *REPENTOGON FOR REPENTANCE+ IS HERE!! (REPENTOGON Launcher Guide)* [Video]. YouTube. <https://www.youtube.com/watch?v=hF4ngfDn364>

GeeksforGeeks. (2025, October 9). *Deep QLearning in Reinforcement learning*. GeeksforGeeks. <https://www.geeksforgeeks.org/deep-learning/deep-q-learning/>